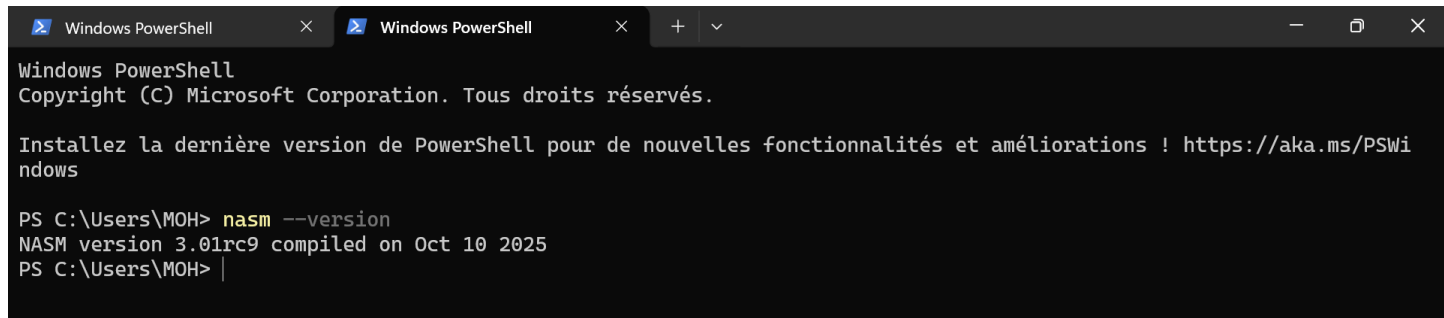
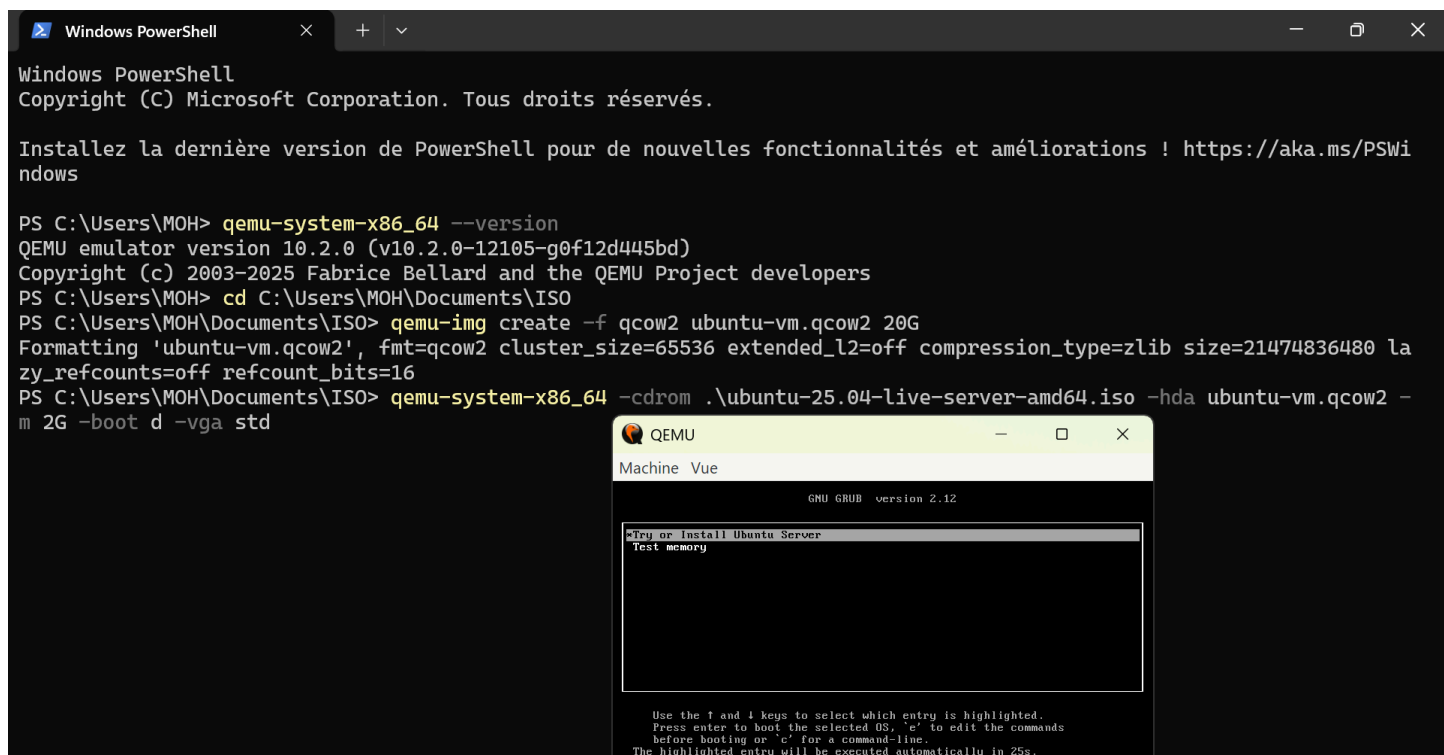
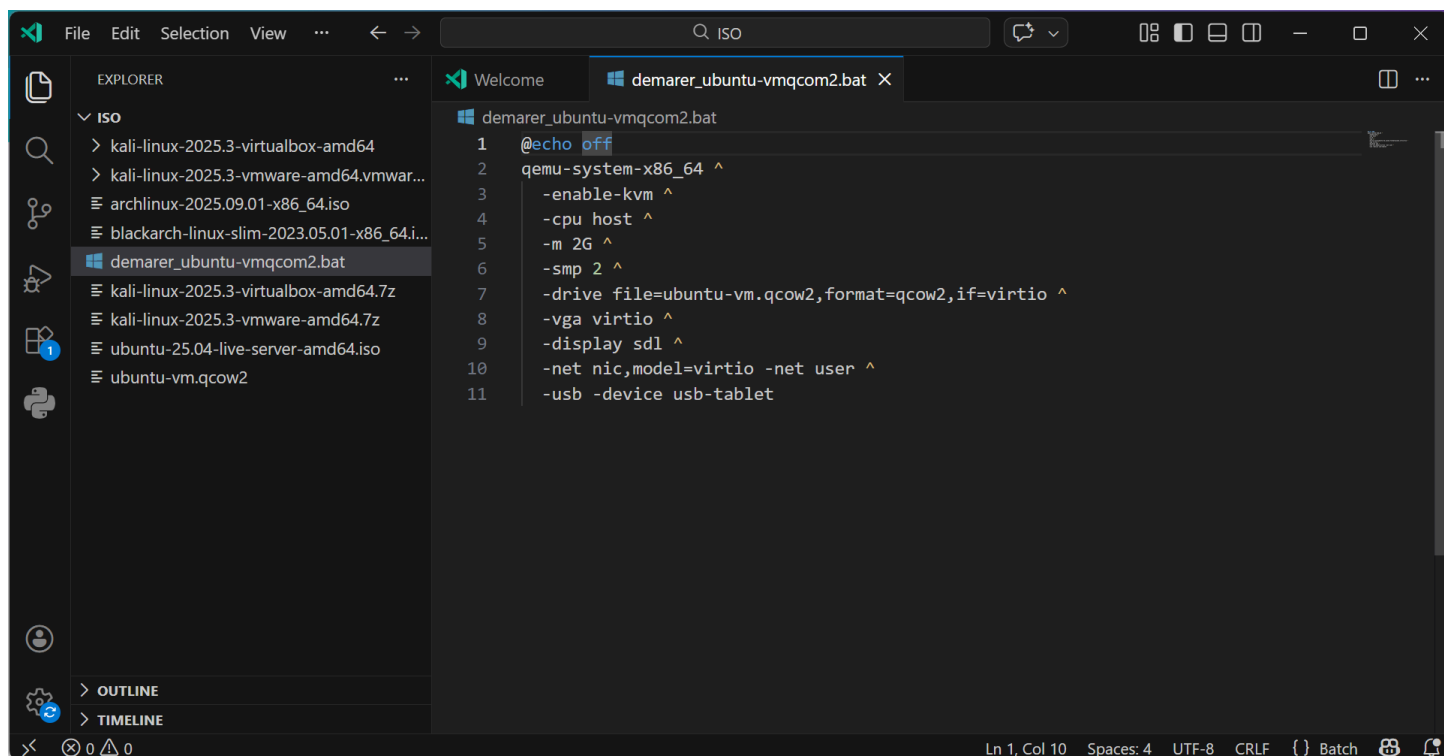




```
Windows PowerShell
PS C:\Users\MOH> gcc --version
gcc.exe (Rev9, Built by MSYS2 project) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

PS C:\Users\MOH> |
```

```
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER$ tree ProjetOS/
```

```
ProjetOS/
├── iso
│   └── boot
│       ├── grub
│       │   ├── menu.lst
│       │   └── stage2_eltorito
│       └── kernel.elf
├── kernel.elf
├── kmain.c
├── kmain.o
├── link.ld
├── loader.o
└── loader.s
```

```
4 directories, 9 files
```

```
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER$ |
```

```
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER/ProjetOS$ tree ../ProjetOS/
```

```
../ProjetOS/
├── iso
│   └── boot
│       ├── grub
│       │   ├── menu.lst
│       │   └── stage2_eltorito
│       └── kernel.elf
├── kmain.c
├── kmain.o
├── link.ld
├── loader.o
└── loader.s
```

```
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER/ProjetOS$ grub-mkrescue -o ../os.iso iso
xorriso 1.5.6 : RockRidge filesystem manipulator, libburnia project.
```

```
Drive current: -outdev 'stdio:../os.iso'
Media current: stdio file, overwriteable
Media status : is blank
Media summary: 0 sessions, 0 data blocks, 0 data, 478g free
Added to ISO image: directory '/'='/tmp/grub.CnGPAu'
xorriso : UPDATE : 588 files added in 1 seconds
Added to ISO image: directory '/'='/mnt/c/Users/MOH/Documents/Lab1_CYBER/ProjetOS/iso'
xorriso : UPDATE : 593 files added in 1 seconds
xorriso : NOTE : Copying to System Area: 512 bytes from file '/usr/lib/grub/i386-pc/boot_hybrid.img'
xorriso : UPDATE : 49.07% done
ISO image produced: 5804 sectors
Written to medium : 5804 sectors at LBA 0
Writing to 'stdio:../os.iso' completed successfully.
```

```
one@DESKTOP-5RV2RM1: /m Windows PowerShell
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER/ProjetOS$ qemu-system-i386 -cdrom ../os.iso -boot d -serial stdio -m 32M
```

QEMU

Machine View

GNU GRUB version 2.12

Minimal BASH-like line editing is supported. For the first word, TAB lists possible command completions. Anywhere else TAB lists possible device or file completions. To enable less(1)-like paging, "set pager=1".

grub>

```
one@DESKTOP-5RV2RM1: /m Windows PowerShell
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER/ProjetOS$ ls
kmain.c kmain.o link.ld loader.o loader.s
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER/ProjetOS$ grub-mkrescue -o ../os.iso iso # depuis Projet OS/
xorriso 1.5.6 : RockRidge filesystem manipulator, libburnia project.

Drive current: -outdev 'stdio'
Media current: stdio file,
Media status : is blank
Media summary: 0 sessions,
Added to ISO image: directory
xorriso : UPDATE : 588
Added to ISO image: directory
xorriso : UPDATE : 593
xorriso : NOTE : Copying to
xorriso : UPDATE : Thank you
ISO image produced: 5806 sectors
Written to medium : 5806 sectors
Writing to 'stdio:../os.iso'

grub> insmod multiboot
grub> multiboot
Possible commands are:
grub> multiboot2
grub> multiboot /b
Possible files are:
boot/ boot.catalog
grub> multiboot /boot/kernel.elf
grub>
```

QEMU - Press Ctrl+Alt+G to release grab

Machine View

GNU GRUB version 2.12

Minimal BASH-like line editing is supported. For the first word, TAB lists possible command completions. Anywhere else TAB lists possible device or file completions. To enable less(1)-like paging, "set pager=1".

grub>

```
one@DESKTOP-5RV2RM1:/mnt/c/Users/MOH/Documents/Lab1_CYBER/ProjetOS$ qemu-system-i386 -cdrom ../os.iso -boot d -serial stdio -m 32M
```

Machine View

Il est vivant !!!! :) :)

GNU GRUB version 2.12

Minimal BASH-like line editing is supported. For the first word, TAB lists possible command completions. Anywhere else TAB lists possible device or file completions. To enable less(1)-like paging, "set pager=1".

grub&gt; insmod multiboot

grub&gt; mu

Possible commands are:

multiboot multiboot2

grub&gt; multiboot /b

Possible files are:

boot/ boot.catalog

grub&gt; multiboot /boot/kernel.elf

grub&gt; boot

-